Report

Of

AI AUTOMATION INTERN : KUAVA DIGITAL

Submitted

To

Ms. Surabhi Purwar

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

By

Roll Number of Student: CS-23411265

Name of Student: SHIVAM SHARMA

Batch: 2026-27

CANDIDATE'S DECLARATION

I **Shivam Sharma**, do hereby declare that the internship report titled " AI AUTOMATION INTERN : KUAVA DIGITAL " has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: Shivam Sharma

Roll No.: CS-23411265

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the KUAVA Digital team for the opportunity to work as an AI Automation Intern and for their guidance throughout the internship period.

I express my gratitude to my parents for their blessings.

Signature

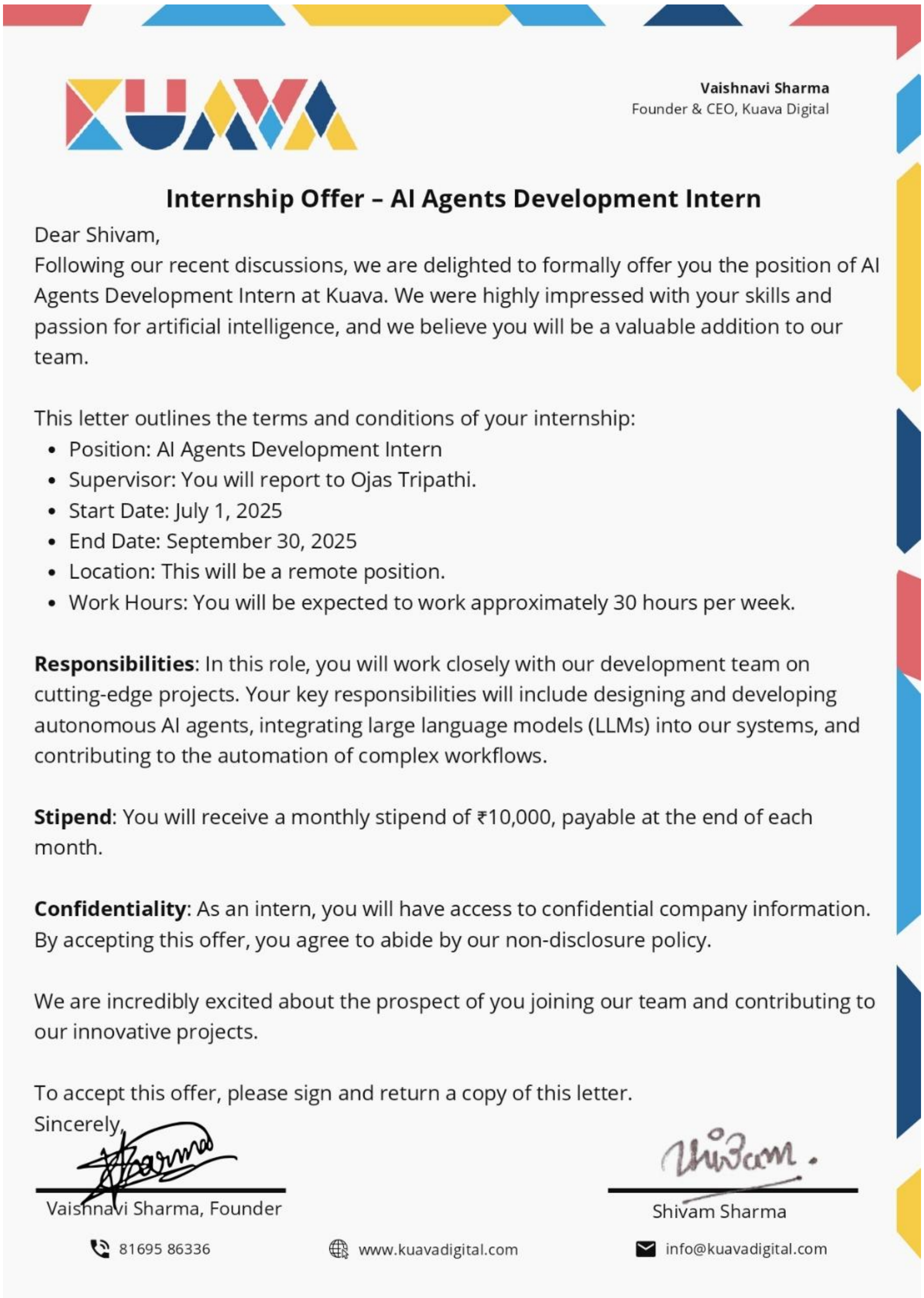
Student Name: Shivam Sharma

Roll No.: CS-23411265

INTERNSHIP COMPLETION CERTIFICATE

The Internship Offer Letter issued by **KUAVA Digital** confirming the position of **AI Agent Developer Intern** is attached below as proof of the internship opportunity. The formal **Internship Completion Certificate** issued by KUAVA Digital upon successful completion of the internship is also attached as supporting documentation.

[Attach: ScaleOn Internship Offer Letter / Completion Certificate here]



KUAVA

Vaishnavi Sharma
Founder & CEO, Kuava Digital

Internship Offer – AI Agents Development Intern

Dear Shivam,

Following our recent discussions, we are delighted to formally offer you the position of AI Agents Development Intern at Kuava. We were highly impressed with your skills and passion for artificial intelligence, and we believe you will be a valuable addition to our team.

This letter outlines the terms and conditions of your internship:

- Position: AI Agents Development Intern
- Supervisor: You will report to Ojas Tripathi.
- Start Date: July 1, 2025
- End Date: September 30, 2025
- Location: This will be a remote position.
- Work Hours: You will be expected to work approximately 30 hours per week.

Responsibilities: In this role, you will work closely with our development team on cutting-edge projects. Your key responsibilities will include designing and developing autonomous AI agents, integrating large language models (LLMs) into our systems, and contributing to the automation of complex workflows.

Stipend: You will receive a monthly stipend of ₹10,000, payable at the end of each month.

Confidentiality: As an intern, you will have access to confidential company information. By accepting this offer, you agree to abide by our non-disclosure policy.

We are incredibly excited about the prospect of you joining our team and contributing to our innovative projects.

To accept this offer, please sign and return a copy of this letter.

Sincerely,

Vaishnavi Sharma, Founder

Shivam Sharma

81695 86336 www.kuavadigital.com info@kuavadigital.com

CERTIFICATE

OF INTERNSHIP

We are pleased to present this Certificate of Completion to:

Shivam Sharma

In recognition of his successful completion of the

AI Agents Development Internship.

During this intensive three-month program, he demonstrated exceptional skill in designing and implementing autonomous AI agents. He actively contributed to projects involving large language models and complex problem-solving frameworks, showcasing a strong aptitude for intelligent automation. Congratulations on this significant achievement in your technological journey. We wish you continued success in your future endeavors.

01/10/2025
Date

Vaishnavi Sharma
Founder

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4. PROJECT DESCRIPTION

4.1 Introduction

During my internship at **KUAVA Digital**, I worked on multiple AI-powered automation projects focused on improving business operations, lead management, customer engagement, and workflow efficiency. The internship provided me with practical exposure to Artificial Intelligence, Large Language Models (LLMs), Python development, workflow automation, databases, APIs, and tools such as Make, n8n, and Vapi AI.

One of the major projects involved developing an **AI-Based Data Extraction Agent** that automatically processed and extracted structured information from emails and stored the relevant data in Google Sheets, reducing manual data-entry efforts and improving accuracy.

I also worked on **Lead Scraping and Cold Email Automation**, where potential leads were collected, organized, and used in automated outreach campaigns. Another automation focused on **meeting scheduling**, where users could book available slots, automatically receive Google Meet links, and get SMS notifications containing their appointment details.

A significant part of my internship involved developing **AI Voice Agents using Vapi AI**. These agents supported real-time voice conversations, appointment booking, inbound and outbound calling, and database-driven interactions. Users could trigger the voice agent directly through a **"Book Appointment" button** on the website navigation bar. The agent could interact with users, collect required information, check availability, and assist with booking appointments.

Additionally, I worked on **outbound AI calling systems**, where the voice agent retrieved relevant information from a database and conducted automated conversations with users. I also contributed to **A/B testing automation**, enabling the organization to test and compare different approaches and improve campaign performance and user engagement.

4.2 Organization Profile

KUAVA Digital is a privately held digital marketing company headquartered in Mumbai, Maharashtra, India. Founded in 2023, the company operates in the marketing services industry and focuses on helping businesses strengthen their digital presence through creative, strategic, and technology-driven solutions.

KUAVA Digital provides a comprehensive range of services, including Social Media Marketing, Digital Marketing, Search Engine Optimization (SEO), Paid Social Media Marketing, Email Marketing, UI/UX Design, Website Development and Design, App Innovation, and Content Creation. The organization combines creativity, innovation, and strategic expertise to develop customized solutions that help brands effectively engage with their target audience and achieve their digital marketing goals.

In addition to its core digital marketing services, KUAVA Digital focuses on adopting modern technologies and innovative approaches to enhance business processes and customer experiences. The company promotes a creative and technology-driven work environment where new ideas and digital solutions are developed to address real-world business challenges.

During my internship at KUAVA Digital, I had the opportunity to work on various AI-powered automation and intelligent agent projects, which provided me with valuable industry exposure and practical experience in applying modern technologies to real-world business applications.

4.3 Problem Statement

Modern digital businesses require efficient automation systems to manage leads, customer communication, appointments, marketing campaigns, and user interactions. At KUAVA Digital, several challenges were identified in managing repetitive and time-consuming business processes manually.

The major challenges included handling unstructured email data, collecting and managing potential leads, performing cold-email outreach, scheduling meetings, sending appointment details through Google Meet links and SMS notifications, and improving customer interaction through AI-powered voice systems. There was also a need for automated A/B testing to improve marketing performance and an intelligent financial agent capable of analyzing and providing insights into the US financial market.

Therefore, there was a need to develop AI-powered and automated solutions for:

1. AI-based Email Data Extraction and Processing.
2. Lead Scraping and Cold Email Automation.
3. Automated Meeting Scheduling, Google Meet Link Generation, and SMS Notifications.
4. AI Voice Agents for Website Interaction, Appointment Booking, and Inbound/Outbound Calling.
5. A/B Testing Automation for improving campaign performance.
6. A US Market Financial Agent for financial data analysis and insights.

These challenges required the integration of Artificial Intelligence, Large Language Models, workflow automation, APIs, databases, and modern AI frameworks to improve efficiency, accuracy, scalability, and customer engagement.

4.4 Project Objectives

The primary objective of the internship was to design and develop AI-powered automation solutions that could improve business efficiency, reduce manual work, enhance customer interaction, and automate repetitive workflows at KUAVA Digital. The internship also aimed to explore the practical implementation of modern Artificial Intelligence technologies and integrate them into real-world business processes.

The key objectives of the projects were:

1. AI-Based Email Data Extraction: To automatically extract structured information from unstructured emails and store it in Google Sheets or databases, reducing manual data entry and improving accuracy.

2. Lead Scraping Automation: To automate the collection, filtering, and organization of potential business leads from relevant sources for efficient lead management.

3. Cold Email Automation: To automate personalized email outreach and improve the efficiency of lead communication and follow-up processes.

4. Meeting Scheduling Automation: To automate appointment booking, slot selection, Google Meet link generation, and SMS notifications containing meeting details.

5. AI Voice Agents: To develop AI-powered voice agents using Vapi AI for real-time conversations, appointment booking, inbound and outbound calling, and database-driven interactions.

6. Website Voice Interaction: To enable users to interact with an AI voice agent through the website, including triggering the agent through the **"Book Appointment"** feature, making the booking process more interactive and accessible.

7. A/B Testing Automation: To automate the testing and comparison of different campaigns or strategies for improving marketing performance, conversion rates

8. US Market Financial Agent: To develop an intelligent AI agent capable of analyzing financial information, processing market-related data, and providing relevant insights related to the US financial market.

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4.5 Scope of the Project

The scope of the internship project involved the design, development, and implementation of multiple AI-powered automation systems aimed at improving business operations, marketing processes, customer communication, and user engagement at KUAVA Digital.

The scope of the projects included:

1. **AI-Based Email Data Extraction:** Development of an intelligent system to extract important structured information from unstructured emails and automatically store it in Google Sheets or databases.

2. **Lead Scraping Automation:** Automation of the process of collecting, filtering, and organizing potential business leads from relevant sources for marketing and outreach activities.

3. **Cold Email Automation:** Development of automated workflows for sending personalized cold emails and managing communication with potential leads.

4. **Meeting Scheduling Automation:** Automation of appointment booking, available slot selection, Google Meet link generation, and SMS notifications containing meeting details.

5. **AI Voice Agents:** Development of AI-powered voice agents using Vapi AI for conducting natural conversations, handling customer queries, booking appointments, and performing inbound and outbound calls.

6. **Website Voice Interaction:** Integration of a voice-based interaction system with the website, allowing users to trigger an AI agent through the **"Book Appointment"** feature and interact through voice.

7. **A/B Testing Automation:** Development of automated systems for testing and comparing different marketing campaigns, strategies, or content to identify better-performing approaches.

8. **US Market Financial Agent:** Development of an AI-powered financial agent capable of processing market-related information and providing useful insights.

4.6 Technologies and Tools Used

During the internship, various AI frameworks, programming languages, automation platforms, APIs, databases, and communication tools were used to develop and implement the different projects.

1. **AI-Based Email Data Extraction**
    - **Technologies Used:** Python, Large Language Models (LLMs), Gmail API, Google Sheets API, LangChain, Make/n8n.
    - Used for extracting structured information from unstructured emails and automatically storing the processed data in Google Sheets or databases.
2. **Lead Scraping Automation**
    - **Technologies Used:** Python, Web Scraping Libraries, APIs, Databases, Make/n8n.
    - Used for collecting, filtering, organizing, and managing potential business leads from different relevant sources.
3. **Cold Email Automation**
    - **Technologies Used:** Python, Email APIs, LLMs, Make/n8n, Databases.
    - Used for generating personalized email content, automating email outreach, and managing communication workflows.
4. **Meeting Scheduling Automation**
    - **Technologies Used:** Google Calendar API, Google Meet Integration, SMS APIs, Make/n8n, APIs.
    - Used for appointment booking, checking available slots, generating meeting links, and sending appointment details through SMS.
5. **AI Voice Agent**

- **Technologies Used:** Vapi AI, LLMs, Speech-to-Text (STT), Text-to-Speech (TTS), APIs, Python.
- Used for developing real-time voice conversations, answering user queries, and providing human-like interactions.

6. **Website Appointment Booking Voice Agent**
   - **Technologies Used:** Vapi AI, Website Integration, JavaScript, APIs, LLMs, Databases.
   - Used to trigger the AI voice agent from the **"Book Appointment"** button and assist users with appointment scheduling.
7. **Inbound and Outbound Voice Calling**
   - **Technologies Used:** Vapi AI, APIs, Databases, LLMs, Python.
   - Used for handling incoming and outgoing calls, retrieving relevant customer information, and conducting context-aware conversations.
8. **A/B Testing Automation**
   - **Technologies Used:** Python, APIs, Databases, Make/n8n, Analytics Tools.
   - Used for automating the comparison and analysis of different campaigns, content, or marketing strategies.
9. **US Market Financial Agent**

- **Technologies Used:** Python, LangChain, LangGraph, CrewAI, RAG, LangSmith, LLMs, Financial APIs, Databases.
- Used for retrieving financial information, analyzing US market-related data, and providing AI-generated insights.

4.7 System Architecture

The following system architectures represent the major automation solutions developed during the internship. Each architecture illustrates the flow of data, AI processing, automation, and integration between different services.

4.7.1 AI-Based Email Data Extraction Automation

Architecture Flow:
Email Source → Email Content Extraction → AI/LLM Processing → Data Structuring and Validation → Google Sheets/Database → Workflow Automation → Process Completion

Component	Role
Email Source	Provides incoming emails containing unstructured business information.
Content Extraction	Extracts email text, metadata, and relevant information for processing.
AI/LLM Processing	Identifies and extracts required structured information from unstructured email content.
Data Structuring and Validation	Organizes the extracted information and checks for incomplete or duplicate data.
Google Sheets/Database	Stores the processed and structured information for further use.
Make/n8n Automation	Manages triggers, routing, integration, and automated workflow execution.

4.7.2 – Lead Scraping Automation

Architecture Flow:

Lead Source Identification → Lead Data Collection → Lead Scraping → Data Cleaning and Validation → Lead Enrichment → Database/Google Sheets Storage → Workflow Automation → Leads Ready for Outreach

Component	Role
Lead Sources	Provides relevant platforms and sources for collecting potential business leads.
Lead Data Collection	Collects information such as names, companies, websites, and contact details.
Lead Scraping	Automatically extracts relevant lead information using scraping tools and APIs.
Data Cleaning and Validation	Removes duplicate, incomplete, or invalid lead records.
Lead Enrichment	Adds additional business or contact information when available.
Database/Google Sheets	Stores organized and validated lead information.
Make/n8n Automation	Controls the automated workflow, triggers, and routing between different stages.
Outreach Ready Leads	Provides validated lead data for cold-email and marketing campaigns.

4.7.3 – Cold Email Automation

Architecture Flow:

Lead Import → Lead Data Processing → AI-Based Email Personalization → Email Scheduling → Automated Email Delivery → Response Tracking → Automated Follow-Up → Campaign Analysis

Component	**Role**
Lead Import	Imports validated lead information from the database or Google Sheets.
Lead Data Processing	Organizes relevant lead information required for personalized communication.
AI/LLM Personalization	Generates personalized and relevant email content based on lead information.
Email Scheduling	Determines the appropriate timing for sending outreach emails.
Automated Email Delivery	Sends emails automatically through the integrated email system.
Response Tracking	Monitors email delivery, responses, and user engagement.
Automated Follow-Up	Sends follow-up messages based on predefined workflow conditions.
Campaign Analysis	Evaluates outreach performance and engagement results.

4.8 Methodology

Email The methodology followed during the internship focused on understanding business requirements, designing AI-powered workflows, integrating different technologies, and developing automated solutions. Each project was developed using a structured approach involving data collection, AI processing, workflow automation, integration, testing, and optimization.

4.8.1 AI-Based Email Data Extraction Methodology

1. **Email Collection and Processing:** Incoming emails were automatically collected and processed to extract relevant content, metadata, and other required information.

2. **AI-Based Information Extraction:** Large Language Models (LLMs) were used to understand unstructured email content and identify important information such as names, contact details, dates, and other relevant fields.

3. **Data Validation and Structuring:** The extracted information was cleaned, structured, and validated to reduce duplicate, incomplete, or inconsistent data.

4. **Storage and Automation:** The processed data was synchronized with Google Sheets or databases, while automation tools were used to manage triggers and workflow execution.

4.8.2 Lead Scraping Automation Methodology

1. **Lead Source Identification:** Relevant platforms and sources were identified to collect potential business leads based on project requirements.

2. **Automated Data Collection:** Lead details such as names, company information, email addresses, websites, and contact details were collected using automated tools and APIs.

3. **Data Cleaning and Enrichment:** The collected information was cleaned to remove duplicate or invalid records, and additional relevant details were added where available.

4.8.3 Cold Email Automation Methodology

1. **Lead Preparation:** Validated leads were imported from the available database or Google Sheets and prepared for the email outreach process.

2. **Personalized Content Generation:** AI and LLM-based technologies were used to generate relevant and personalized email content based on lead information.

3. **Automated Email Delivery:** Emails were scheduled and sent automatically through integrated email and workflow automation systems.

4. **Follow-Up and Tracking:** Email responses and engagement were monitored, while automated follow-ups were triggered based on predefined conditions.

4.8.4 Meeting Scheduling Automation Methodology

1. **Appointment Request and Slot Selection:** The system received appointment requests and checked available time slots, allowing users to select a suitable meeting time.

2. **Meeting Creation:** Once the appointment was confirmed, the meeting was scheduled automatically and a Google Meet link was generated.

3. **Notification System:** Meeting details, including date, time, and the Google Meet link, were automatically sent to users through SMS notifications.

4. **Workflow Automation:** APIs and automation platforms were used to connect all components and automate the complete scheduling process.

4.8.5 AI Voice Agent Methodology

1. **Voice Input Processing:** The AI Voice Agent processed user voice input in real time to enable natural and interactive conversations.

2. **AI Response Generation:** Large Language Models were used to understand user queries, maintain context, and generate intelligent responses.

3. **Real-Time Voice Interaction:** AI-generated responses were converted into natural speech, enabling smooth communication between users and the voice agent.

4.8.6 Website Appointment Booking Voice Agent Methodology

1. **Website Integration:** The AI voice agent was integrated with the website and could be triggered through the **“Book Appointment”** button.
2. **User Interaction:** The agent communicated with users through voice and collected the necessary appointment-related information.
3. **Appointment Processing:** The collected information was processed and connected to the appointment scheduling workflow.
4. **Automated Confirmation:** The appointment details were passed through the automation system to complete scheduling and confirmation-related actions.

4.8.7 Inbound and Outbound Voice Agent Methodology

1. **Inbound Call Handling:** The AI agent handled incoming calls, understood user queries, and provided real-time conversational responses.
2. **Outbound Calling:** Relevant information was retrieved from the database before initiating outbound calls to enable personalized conversations.
3. **Context-Aware Interaction:** Database information was provided as context to the AI agent to generate relevant responses during conversations.
4. **Workflow Integration:** Information collected during calls was stored and integrated with the required databases or automation workflows.

4.8.8 A/B Testing Automation Methodology

1. **Variation Creation:** Different versions of campaigns, content, or strategies were prepared for comparison.
2. **Automated Distribution:** The variations were distributed through the required workflows to conduct the testing process efficiently.
3. **Performance Data Collection:** Relevant performance metrics such as engagement, responses, and conversions were collected automatically.
4. **Result Analysis:** The results were compared to identify better-performing variations and support data-driven marketing decisions.

4.8.9 US Market Financial Agent Methodology

1. **Financial Data Retrieval:** Relevant US market and financial information was retrieved from available sources and financial APIs.

2. **AI-Based Analysis:** Large Language Models were used to process financial information, understand user queries, and generate meaningful insights.

3. **Agent Framework Integration:** LangChain, LangGraph, CrewAI, and RAG were used to develop intelligent workflows, manage agent interactions, and retrieve relevant contextual information.

4.9 Expected Output

The expected output of the internship projects was the successful development of AI-powered automation systems capable of improving business efficiency, reducing manual effort, and enhancing customer interaction. The expected outcomes included:

1. **Automated Data Processing:** Structured information extracted automatically from unstructured emails and stored in Google Sheets or databases with reduced manual effort.

2. **Improved Lead Management and Outreach:** Automated lead scraping, organization, and cold-email workflows to support efficient lead generation and communication.

3. **Automated Appointment Management:** A complete scheduling workflow capable of handling slot selection, appointment booking, Google Meet link generation, and SMS notifications.

4. **Intelligent Voice Interaction:** AI-powered voice agents capable of handling real-time conversations, appointment booking, inbound and outbound calls, and database-driven interactions.

5. **Improved Marketing Performance:** Automated A/B testing workflows capable of comparing different campaigns and identifying better-performing strategies.

6. **Financial Market Insights:** A US Market Financial Agent capable of processing relevant financial information and providing useful market-related insights.

7. **Scalable AI Solutions:** Integration of modern technologies such as LLMs, LangChain, LangGraph, CrewAI, RAG, LangSmith, Vapi AI, Make, n8n, APIs, and databases to create scalable and efficient automation systems.

4.10 Certificates of Completion and Other Communication Mail Proof

The **KUAVA Digital Internship Offer Letter and Internship Completion Certificate** are attached to this report as supporting documents and proof of the successful completion of the internship.

Vaishnavi Sharma
Founder & CEO, Kuava Digital

Internship Offer – AI Agents Development Intern

Dear Shivam,

Following our recent discussions, we are delighted to formally offer you the position of AI Agents Development Intern at Kuava. We were highly impressed with your skills and passion for artificial intelligence, and we believe you will be a valuable addition to our team.

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- Location: This will be a remote position.
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We are incredibly excited about the prospect of you joining our team and contributing to our innovative projects.

To accept this offer, please sign and return a copy of this letter.

Sincerely,

Sharma

Vaishnavi Sharma, Founder

Shivam.

Shivam Sharma

81695 86336 www.kuavadigital.com info@kuavadigital.com

CERTIFICATE

OF INTERNSHIP

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In recognition of his successful completion of the

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01/10/2025
Date

Vaishnavi Sharma
Founder

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